Even Hand: Design Document

A Grading-Consistency Audit for IIT Madras Courses

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`https://github.com/Zayaan3019/Even_Hand`

1 What changed since the proposal

Scope was approved by Prof.Aslam unchanged, so this revision is driven by what we learned after 1 September.

(a) The Institute prescribes no marks-to-grades rule. The *B.Tech Ordinances* (2015) fix grades and points (R.21.1), make marking absolute (R.19.2) and leave letters to the Class Committee (R.22.1, R.4.4(d)); no mean, standard deviation, top share or cut-off appears. So the bands our proposal called institute policy are departmental convention, and the rule is per-course configuration.

(b) Grade points are unevenly spaced (S=10, A=9, B=8, C=7, D=6, **E=4**, U=0; a U in a core course must be repeated, R.25.1), so the shortlist ranks by expected grade-point change, $P(\text{grade changes}) \times \Delta$ points, not by probability alone.

(c) Severity harms a student only through *differential exposure* (§2.3). When each question's marks of record come from one assistant, a harsh assistant is indistinguishable from a harder question *and moves nobody relative to anybody*. Pure question-wise marking — the proposal's "failure" case, and our stakeholder's arrangement — is therefore *safe by design*, and the rotation we meant to recommend can *create* the harm it then has to measure. The plan changes accordingly (§4).

2 Architecture

2.1 One pipeline, nine modules

A course enters at `eh-ingest` and leaves as a report from `eh-report`; `./mvnw verify` builds and tests all of it. Dependencies flow strictly downward, and `eh-estimate`'s POM does not offer Spring, JPA or SQLite, so "the estimator is tested without a database" is a property of the build graph. ArchUnit asserts the same rules; `.github/CODEOWNERS` maps each module to its owner. †: signature fixed here, implementation planned.

Module	Stage	Exports across its boundary	Owner
`eh-core`	IR, seam	`MarkingDesign` (no score; carries `AllocationEvidence`); sealed `AuditVerdict` with its `Exposure` and `IdentifiabilityCertificate`; `AnchoredDesign.of(design, verdict, anchoring)`; `FittedModel`; `CourseProfile`, every field `Provenanced`	Zayaan
`eh-numerics`	arithmetic	`ExactRowSpace` (integer rank, row-space membership, null basis); `SymmetricEigen`; Cholesky†; seeded RNG†	Zayaan
`eh-ingest`	parser	`CsvIngest.read(csv, pseudonymiser)` → responses, design, line-numbered diagnostics; `KeyFile.seal`, `KeyFile.open`	Zayaan
`eh-design`	analysis	`IdentifiabilityAudit.audit(design, weights)` → `AuditVerdict`; `DriftAudit.audit(design)`; `DesignProfiler.profile(design)`	Zayaan
`eh-estimate`	measurement	`Estimator.fit(anchoredDesign, responses)` → `FittedModel`†; `NullTest`†; `UncertaintyEngine`†	Zayaan
`eh-linking`	fix-it hint	`LinkingDesign.propose(design, exposure, targetSE)` → ranked changes†	Pruthviraj
`eh-outcome`	consequence	sealed `GradingRule`†; `Outcome.of(model, profile)` → transfers and a sealed `Shortlist`, `Emitted` or `Refused`†	Pruthviraj
`eh-report`	narration	Instructor report, one-page Class Committee sheet, `RunManifest`†	Pruthviraj
`eh-app`	runtime	Spring Boot and Thymeleaf on 127.0.0.1; JPA/SQLite run store; CLI†	Pruthviraj

The seam. Everything the measurement stage knows leaves it in one `FittedModel` (estimates, covariance on the identified subspace, anchoring, verdict); nothing downstream reaches back, so the estimator can be

replaced without changing what the instructor is told. Its only entry point takes an `AnchoredDesign`, whose single factory refuses a NOT-ESTIMABLE verdict or a certificate that is not true of the design in hand.

2.2 Trust invariants

Each is enforced by a type, the build graph or a named test — never by convention.

I1. **No estimate without an audit of *this* design** (`AnchoredDesign`), checked at use because the audit runs weeks before the marks. The certificate is verified, not trusted: the rank of the information matrix modulo two primes bounds its claimed rank from below, and a null basis that every marking annihilates bounds it from above.

I2. **No corrected mark ever becomes a grade.** Counterfactual marks stay inside `eh-outcome`; the only per-student output is a recommendation to re-check a script, and a grade changes only by a human re-mark (ArchUnit: no export path).

I3. **Refusal is a value**: NOT-ESTIMABLE stops estimation, and a shortlist that fails its gate is a `Refused` value carrying its reason.

I4. **Every figure carries its interval, every input its provenance** (measured, declared or defaulted).

I5. **No identifier on disk except in the key file, which is sealed first; no pseudonym encodes order** (labels in keyed-digest order).

I6. **Nothing leaves the machine**: loopback bind; ArchUnit forbids HTTP clients and sockets.

I7. **Per-assistant figures reach only the instructor**; the committee sheet is aggregate.

I8. **Reports reproduce byte-for-byte** from their manifest and key, using a seeded RNG, sorted iteration and `StrictMath`.

2.3 The audit: who could have been harmed, and can it be measured?

With $\eta_{nij} = \theta_n - \delta_i - \gamma_j$ (ability, difficulty, severity), the audit answers three questions from the allocation alone. **It reads no marks**, so the instructor has the answers before sharing any data; this is what we bring the stakeholder in week one.

1. Exposure: could severity have moved anyone? Let E_{nj} be the share of student n's composite (question maxima $\times$ assessment weights, marks of record only) awarded by assistant j. With one marker of record per question, each question's expected mark rises with ability under the same assistant for everyone, so the expected composite rises with ability *whatever the severities*: rank-based grades are unchanged in expectation, and sigma bands move only as they would for a harder paper. In general the harm is, to first order, $\sum_j (E_{nj} - \bar{E}_j)\, s_{nj}\gamma_j$ with s the local marks-per-logit slope: zero for exposure-balanced rotation, largest for script-wise piles. Summed over assessments, it answers the proposal's allocation-regime question before anything is estimated.

2. Estimability: can what matters be separated? For each pair of assistants we test membership of the design's row space by fraction-free integer elimination — a fact about the arrangement, not a floating-point threshold (Facets' graph heuristic is documented to misreport "exotic forms of connectedness"). A harm contrast $(E_n - E_m)\gamma$ is then estimable exactly when every student carries the same weight inside every group of mutually comparable assistants. Script-wise and question-wise marking are *equally* rank-deficient (excess null dimension $J-1$ beyond the two that anchoring removes) but differ in kind. Question-wise, severity is confounded with difficulty and nothing about students separates them. Script-wise, it is absorbed into the pile's abilities, and is recoverable if the piles were alike. Re-asking against a design whose student columns are collapsed to one intercept tells the two apart — and that collapsed design is exactly the fixed-effect part of the estimator's random-ability model (§2.4), so the audit certifies what the estimator will assume. Random and roll-number splits are the same design, so the audit reports evidence instead: whether piles are contiguous in roll number or follow its department code, computed at ingest from the real roll numbers and kept only as counts. A 400-student course audits in about a second.

3. Drift: does a late script lose? The same test, with marking position as a column, exposes a trap. If every marker works through the stack in the same (usually roll-number) order, the same students are late on *every* question, so the harm accumulates, yet drift is identified only by assuming stack order is unrelated to ability. Having alternate markers reverse the stack identifies drift outright and balances its exposure, at no cost. Under our stakeholder's arrangement, drift, not severity, is the harm that can exist.

The verdict names the groups of mutually comparable assistants — a partition, so two equal-sized groups are both reported — summarised as ESTIMABLE, UNDER-EXCHANGEABILITY, PARTIAL or NOT-ESTIMABLE, and every verdict carries its exposure. Without differential exposure the course is **safe** whatever the verdict, and `EH003` is a note, not an error. With it, the harm is **correctable** if estimable (stated as resting on alike piles when under exchangeability) and otherwise **harmful and unmeasurable**: the tool refuses and names the repair.

What changes in the advice. The linking design optimises, in order: (1) no differential exposure — one marker of record per question, or rotation only in exposure-balanced form (rotating over questions worth 5/10/15/10/10 marks leaves up to 5 of a student's 50 marks with a different mix of assistants from the class's); (2) a small double-marked sample, which measures inconsistency and anchors drift; (3) rotation for per-assistant feedback, only if the instructor wants it. No within-course arrangement identifies the team's *collective* severity; under an absolute rule that needs the instructor to re-mark a sample, and the report says so.

2.4 Estimation and uncertainty

Model. Many-facet Rasch, fitted by *marginal* maximum likelihood (EM over Gauss–Hermite quadrature, abilities a normal random effect; joint ML only for start values). Joint ML was dropped as the estimator: its incidental-parameter bias needs a correction Facets leaves manual; it offers no shrinkage at 60–150 students; and it cannot see that piles differ by chance — at 30 scripts per pile and an ability SD of 1 logit, pile means differ with SD 0.26 logits, two-thirds of the proposal's own 0.4-logit example — so its intervals under exchangeability would be confidently wrong. Scores are modelled in the steps ingest reads; long scales use Andrich's two-parameter dispersion thresholds, so a 20-mark question costs two parameters, not twenty. Severity is per (assistant, session), with pooling tested by likelihood ratio. Erratic marking and central tendency are reported, never folded into severity, and each question's effective weight in the ranking is shown against its nominal weight.

Uncertainty. One engine pushes parameter draws (parametric bootstrap for headline figures) through the composite (R.19.1), boundaries, grades and transfers, so every interval comes from one tested component. **The no-effect test preserves the design**: it simulates the fitted no-severity model on the *same* allocation and refits. Permuting grader labels, as the proposal planned, would turn a question-wise design into a crossed one and test the wrong null. The negative control puts the real allocation's labels on an auto-scored component.

2.5 Consequence, and privacy

`GradingRule` is sealed (sigma-band, top-fraction, absolute, instructor-band; explicit `sBandPolicy`). Counterfactual marks are referenced to the panel-average assistant, and boundaries are recomputed on them. Hand-drawn bands have no knowable counterfactual, so we report the range between holding the thresholds and holding the grade counts. The shortlist gate is pre-registered in week 4:

- a shortlist is emitted only if the no-effect test rejects at 5%;
- each row needs $P(\text{grade changes}) \geq 0.2$, and there are at most 20 rows, ranked by expected grade-point change, with the expected number of false flags printed;
- rows run in both directions, and under a fixed share each promotion names the script it displaces;
- every row carries the R.24.1 revision deadline.

Privacy and security. Threats: a lost laptop, re-identification, appraisal misuse, exfiltration. Identifiers become HMAC-SHA-256 digests under a course-scoped salt, so a course's assessments link but a new offering does not. Review found two gaps, both now fixed. Labels were issued in file order, so in a roll-sorted export "Student 1" was the lowest roll number; they now follow keyed-digest order. And the key file, the only re-identification table, was plain text; it now carries the course salt and is sealed with AES-256-GCM under a passphrase (PBKDF2, 600,000 iterations, header authenticated; JDK only) before it is written. Pseudonyms do little for assistants where the allocation is known ("the Q3 marker" is a name), so per-assistant figures go only to the instructor. Minimal fields, local processing and deletion on request follow the data-sharing note and the DPDP Act, 2023.

3 Test plan

At least one named test per module. Test dependencies are test-scope only, so no third-party numerics reach production. **104 tests pass today**, one of them a property checked on 150 random allocations, all in the four implemented modules; the other five have the signatures above and no implementation yet, so their tests are *planned*.

Defects found in review, now fixed under regression tests: pseudonyms revealed file order; the roll-number evidence read row order, so an export sorted by assistant looked like a roll-number split; half marks failed to parse; the mark of record was undefined; equal-sized groups of assistants were reported as one reachable and one blocked; the eigenvalue cross-check covered one unit matrix; and Q1 of a quiz and Q1 of the end-semester exam were merged into one question.

Module	**Tests** (PASSING today; *planned*)
`eh-core`	PASSING: a certificate is bound to its design, and one claiming more identification, or more confounding, than the marking supports is rejected; `AnchoredDesign` refuses NOT-ESTIMABLE; a verdict cannot over- or under-claim its certificate; the fingerprint ignores row order but not marking order or which mark counts
`eh-numerics`	PASSING: rank vs. hand-computed matrices; duplicate rows never inflate rank; exact membership; null vectors annihilated; eigenvalue count equals exact rank. *Planned*: Cholesky and the seeded RNG
`eh-ingest`	PASSING: **no roll number or assistant name survives ingest**; labels independent of row order; pile evidence from roll numbers, not rows, with department codes; half and quarter marks; the mark of record by role column, first mark or average; questions scoped to their assessment; the key sealed and tamper-evident, the course salt restored; no maximum means refusal; a totals column caught; bad rows reported by line
`eh-design`	PASSING: the scenario matrix (Appendix A) *including refusals*, with exposure and drift; exposure as a property of 150 random allocations; `EH003` a note only when harmless; equal-sized groups both reported; the drift cases; on every scenario, exact and eigenvalue null dimensions agree and the certificate survives re-verification; the profiler recovers the arrangement unaided
`eh-estimate`	*Planned*: parameter recovery; **95% coverage**, including script-wise designs with abilities redrawn each replication; the no-effect test has 5% size; partial credit reduces to dichotomous at m=1; anchoring invariance; monotone EM likelihood; **negative control** inside the null
`eh-linking`	*Planned*: connectivity repair minimal ($c-1$ links); an exposure-balanced rotation found when one exists; never proposes more differential exposure than the current design
`eh-outcome`	*Planned*: rules vs. hand-computed cohorts; fixed-share exchange (#promoted = #displaced) as a property; instructor-band range brackets both limits; shortlist symmetric and refused in $\geq$95% of no-effect simulations
`eh-report`	*Planned*: no key-file name in any output; no per-assistant figure on the committee sheet; R.24.1 date; golden snapshots
`eh-app`	*Planned*: loopback bind; no-network rule; no export path from counterfactual marks; end-to-end over HTTP; full run at N=150 under 60 s

4 Revised milestone plan

The week-by-week plan, followed by the revisions behind it.

Wk	Dates	Zayaan	Pruthviraj
W0	10–11 Sep	DONE: build, seam, exact audit, generator, profiler, ingest, review fixes; 104 tests	Review of week-0 commits
W1	12–18 Sep	Real allocation → exposure, verdict, drift and SE forecast to the stakeholder	`GradingRule` family and composite on a synthetic fit; second course requested
W2	19–25 Sep	Marginal ML (binary), standard errors, recovery test	Transfers; shortlist gate; instructor-band range
W3	26 Sep–2 Oct	Partial credit, dispersion thresholds, uncertainty engine; gate	CLI end to end on a stub fit; committee sheet; golden tests
W4	3–8 Oct	Design-preserving null; coverage study; drift if order known	Thymeleaf views; run store; pre-registration (both)
Mid-demo 9 Oct: the stakeholder's course end to end — exposure, verdict, estimates or refusal, grades under the confirmed rule; gap list			
W5	10–16 Oct	Sessions, rater diagnostics, effective weights	Variance share; transfers with intervals; shortlist on real data
W6	17–23 Oct	Negative control; second course fitted	Exposure-balanced linking optimiser; second-course report
W7	24–30 Oct	Hardening; performance guard	Blind re-grade issued (masked scans); handoff pack and installer
W8	31 Oct–5 Nov	Re-grade analysis; stakeholder dry-run; cross-module viva preparation (both)	
Final submission and viva 6 Nov			

R1. **The audit moves ahead of the marks**, and already gives three answers from the allocation alone: exposure, estimability and drift estimability. Week 1 runs it on the real one.

R2. **A prediction replaces a risk.** If the allocation table confirms one marker of record per question, we pre-register about zero severity-driven transfers on the first course. Drift joins the core wherever marking order is recorded or can be declared.

R3. **The second real course is requested in week 1, not week 7**, preferably script-wise, where transfers

are expected.

R4. **Precision is the binding risk.** The design-only standard-error forecast gates the shortlist; the mid-demo criterion is "estimated, or refused with the repair named".

R5. **Marginal ML replaces JMLE plus manual bias correction.** Plan B is JMLE with pile means constrained and pile variance added analytically, behind the same seam, chosen at the week-3 gate.

R6. **Institutional fit**: configurable cut-off rule (§1(a)), composites (R.19.1), and outputs timed to the Class Committee and the R.24.1 window.

R7. **Workload rebalanced.** Week 0's commits are one member's, and we say so. The seam lets `eh-outcome` start on a synthetic fit, so Pruthviraj leads a deliverable every week and both of us commit weekly.

R8. **The cut line has dates**: question diagnostics on 2 Oct, the linking optimiser on 23 Oct, drift only if no marking order exists. The audit, exposure analysis, estimator, propagation and blind re-grade are never cut; the re-grade's pre-registration is committed in week 4.

5 Risks and plan B

1. **The first course shows no severity harm.** Now expected and pre-registered (R2); a correct "safe by design" is a finding. Transfers are shown on the second course or, failing that, on the stakeholder's own allocation with injected, known severities, labelled synthetic.
2. **Nobody recorded who marked what**, as is usual with paper scripts. Intake will accept the allocation as rules ("TA 2: Q3–Q4, all students"; "TA 1: all questions, CE23B001–060"), with a template to use before the next exam is marked.
3. **Marking order was not recorded.** The instructor may declare roll-number order (flagged DECLARED); recording it at the end-semester exam costs nothing.
4. **Intervals are too wide, or marginal ML slips.** We report the intervals and refuse the shortlist, as a type; the estimator falls back to plan B (R5) at the week-3 gate.
5. **No effect is detected.** We report it, with the effect we can rule out and the sample needed; we do not shop for another course.
6. **An assistant's standing is harmed.** Per-assistant figures go to the instructor only; erratic marking is reported apart from harsh; estimates describe a marking session, not a person.
7. **The blind re-grade is not blind**: returned scripts carry the assistant's ink, so scans are masked before they go back (else ten flagged and ten control scripts, else a reported gap). The test is directional, with its power pre-registered.

Appendix A: the scenario matrix

Output of `AuditReport`, run by CI on every commit; every column is computed by the audit and asserted in the tests, each answer worked out on paper first. Exc.: null dimension beyond the two anchoring removes; —: marking order not recorded. Question maxima are equal except where stated. No row uses a mark.

Arrangement	Exc.	Severity	Exposure	Drift	Headline
Question-wise + double-marking	0	ESTIMABLE	none	—	Safe; the stakeholder's course
Question-wise, marker rotated	0	ESTIMABLE	none	—	Safe
same, maxima 5/10/15/10/10	0	ESTIMABLE	estimable	—	Rotation-made; correctable
Script-wise, random piles	3	UNDER-EXCH.	under-exch.	—	Correctable if the piles were alike
Script-wise, roll-number blocks	3	UNDER-EXCH.	under-exch.	—	Same; the assumption is doubtful
Question-wise, no overlap	4	NOT-ESTIMABLE	none	—	Safe by design
Question-wise, one stack order	4	NOT-ESTIMABLE	none	under-exch.	Safe; drift rests on stack order
alternate stacks reversed	4	NOT-ESTIMABLE	none	estimable	Safe; drift measurable
Mixed across assessments	2	PARTIAL (2+1+1)	estimable	—	Correctable
Single marker	0	NOT-ESTIMABLE	none	—	No grader facet
Two disconnected islands	4	PARTIAL (2+2)	not estimable	—	Harmful; unmeasurable
Islands + one bridging mark	3	UNDER-EXCH.	under-exch.	—	Correctable if the piles were alike